Overview LNG Markets & Trading

Day Two

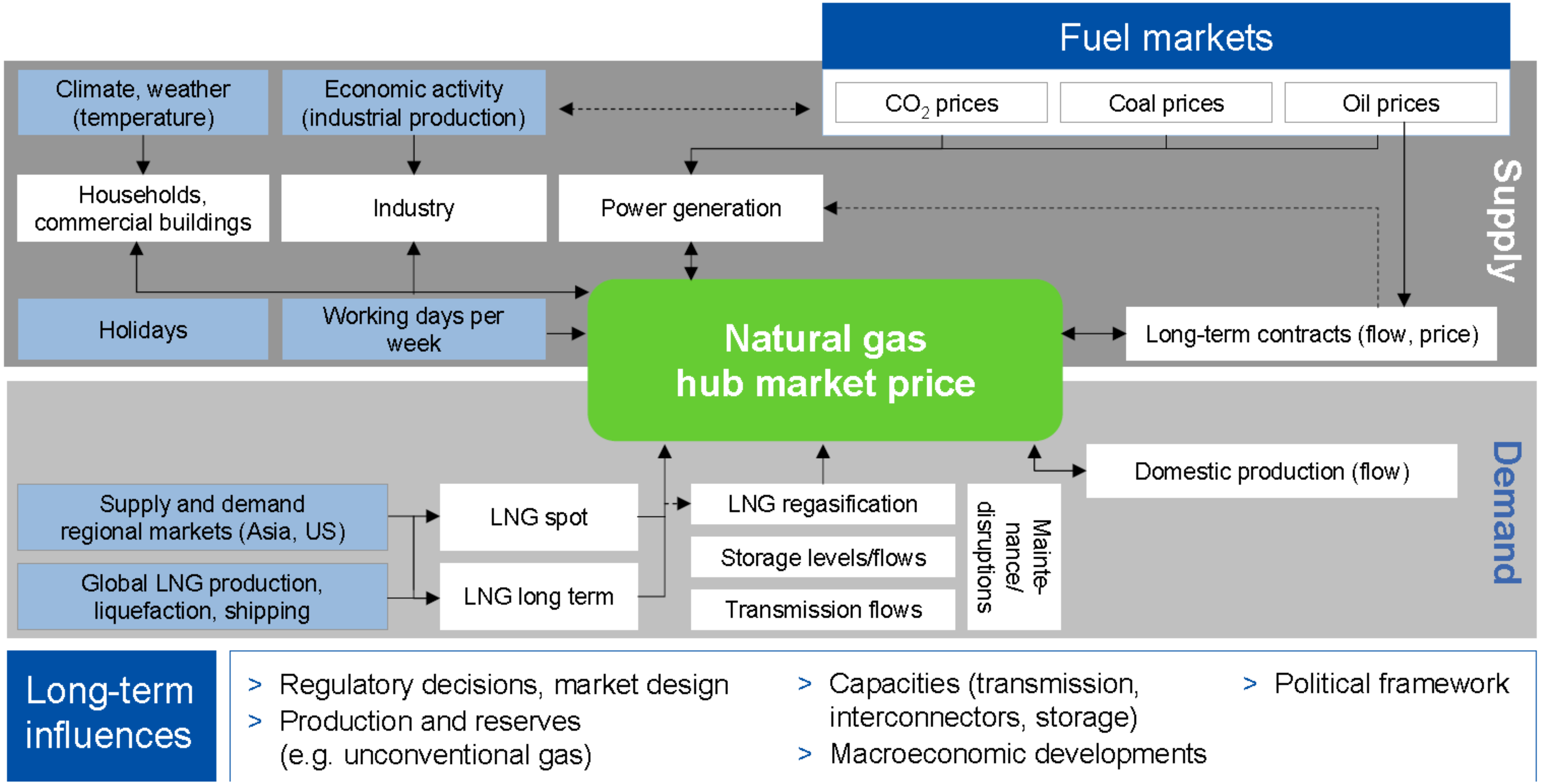
Program Today

- Latest developments
- European Gas Market in short
- Pricing Dynamics
- LNG Shipping
- LNG Contracting

Latest Developments

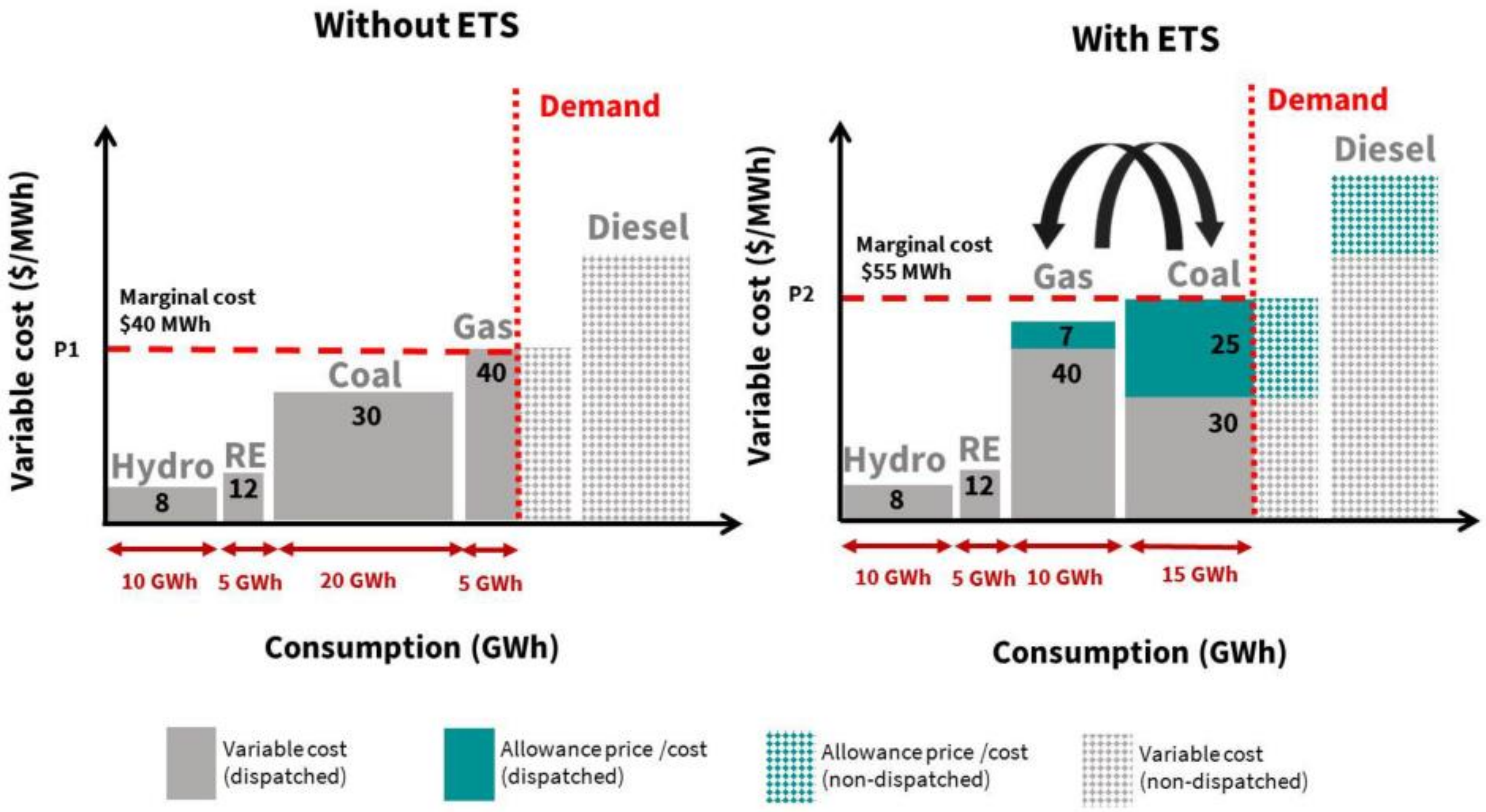
Case European Gas Markets

Drivers European NG Market



ICE

Impact Carbon on Power Price



Pricing Dynamics

LNG Important Price Benchmarks

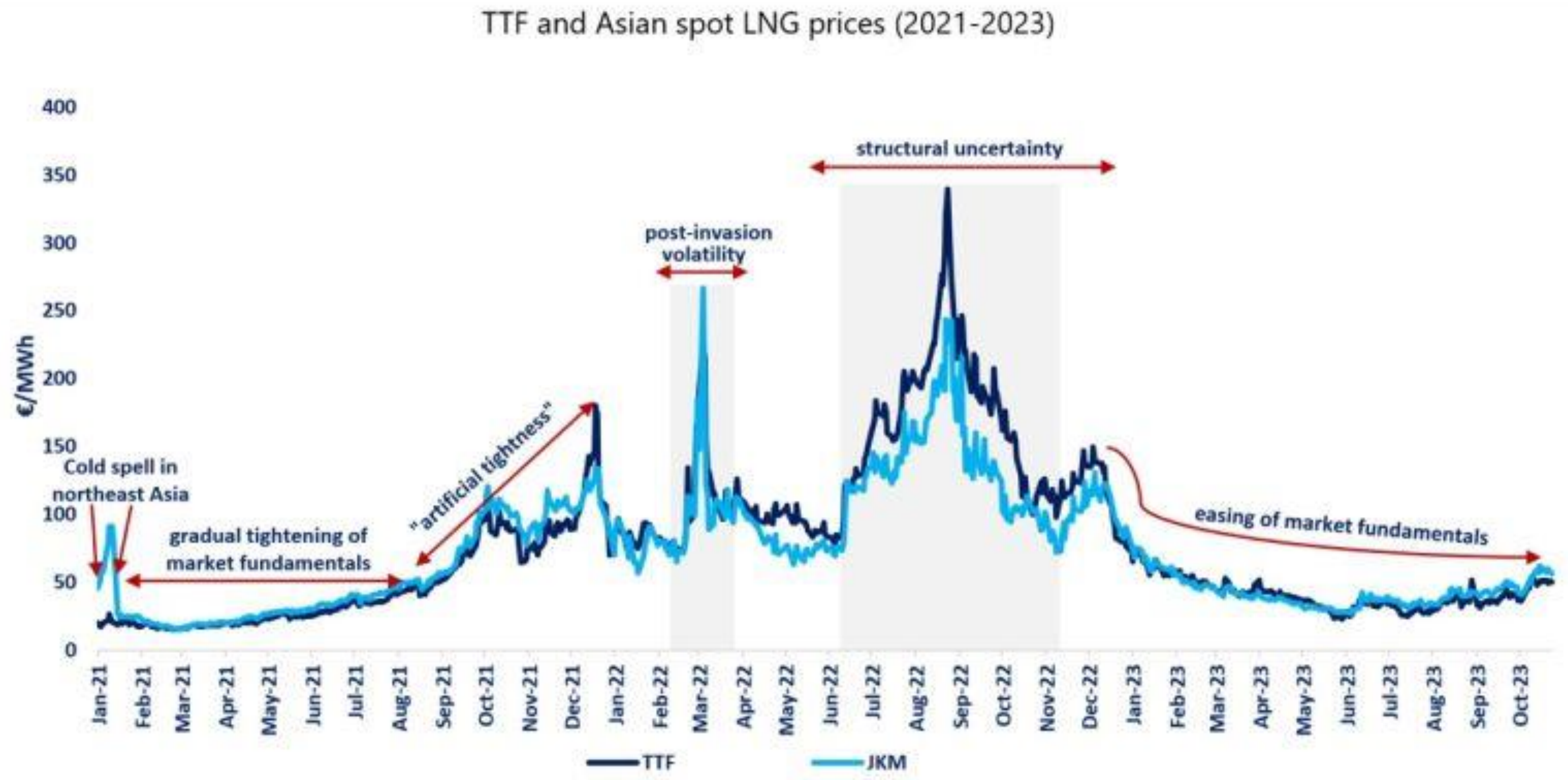
Benchmark	Last close	Daily change %	Weekly change %	Weekly avg.	Last 5 sessions
TTF	$ 14.60	-3.3%	*3.4%*	*$ 14.53*	
JKM	$ 17.19	-0.4%	*-1.5%*	*$ 17.31*	
Henry Hub	$ 3.03	-0.3%	*-7.1%*	*$ 3.12*	
NWE LNG	$ 15.01	-0.8%	*-0.3%*	*$ 15.10*	
Oil-indexed LNG	$ 12.08	0.0%	*0.0%*	*$ 12.08*	
Coal (API2)	$ 4.51	-3.0%	*0.7%*	*$ 4.52*	

Date of last close: *10/11/2023*

All monetary values in $/MMBtu. Oil-indexed LNG is month avg JCC @ 12.5% slope + $0.5 constant.

Global Benchmarks

LNG Spot Prices



Pricing Structures

- Gas hub–linked pricing
- Index switching formulas with hybrids of minimums and caps
- Volume flexibility options
  - Cargo-size tolerance
  - Seasonal swings
  - Carry-forward options
- Tailored credit and payments terms
  - Secured payments on power offtakes
  - More limited guarantees

Pricing Structures

Oil indexed

- Slope * Brent Crude + fixed part
- 3 -6 months lag

- Example:
  - 13 % slope against Brent crude
  - Brent crude price $70/ bbl
  - LNG price = 0.13 x 70 =$ 9.10/ mmbtu

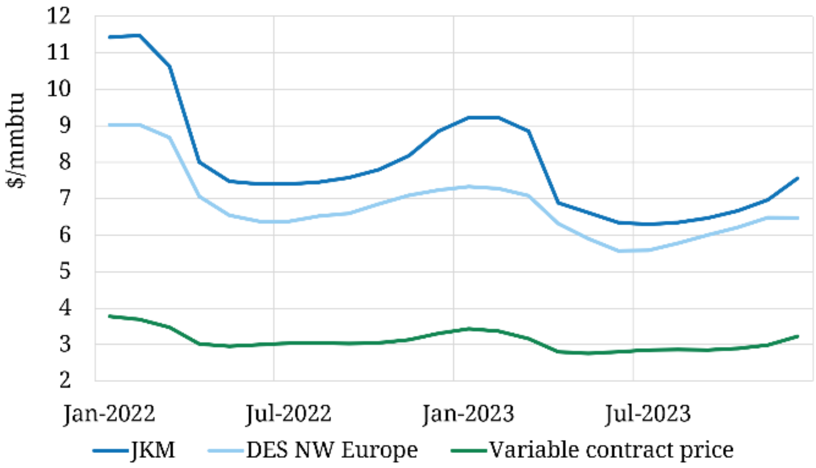
Pricing Structures

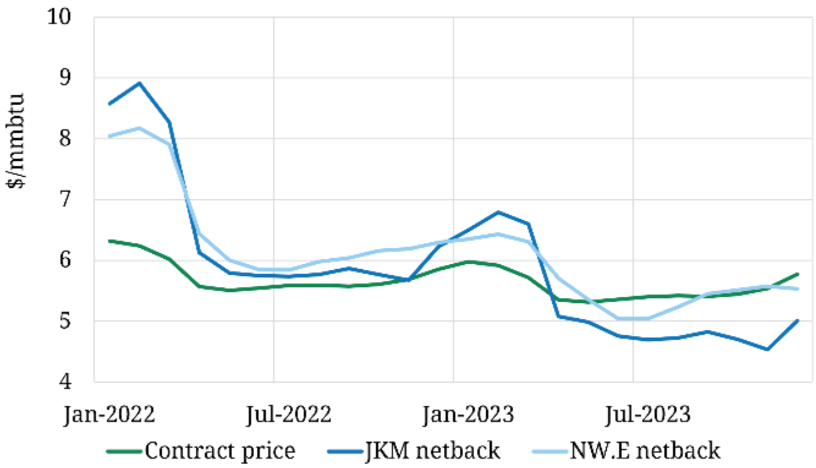
US FOB export price

- 115% * Henry Hub price + fixed liquefaction fee
- Variable part
  - SRMC of exported gas
  - App. half premium covers actual cost of feedgas
- Fixed part
  - Cover the costs of liquefaction terminal development
  - $ 2.25 - $ 3 /mmbtu
- Important from a total supply cost perspective
  - Once export contract is signed, this cost is sunk
  - Largely irrelevant in influencing commercial decisions around the flow of gas

Case Study US Tolling Contracts

Component	Value
Term	Jan 2022- Dec 2023
ACQ	3,600,000 MMBTU per cargo
Cargoes	~ 6 per year rateable, offtake every second month
Fixed tolling fee	2.55 $/mmbtu (payable even if cargo cancelled)
Variable price terms	115% Henry Hub
Cancellation terms	15th day of the second month ahead of delivery (M-2)





Pricing Structures

US FOB export price

- US export contract flow decisions driven by two factors:

1. **Variable cost (or SRMC) of US export contracts**
   - Henry Hub + costs to get LNG onto a vessel
   - Primarily liquefaction

2. **Netback global spot price signals**
   - Market value for exported gas, adjusted for appropriate shipping costs

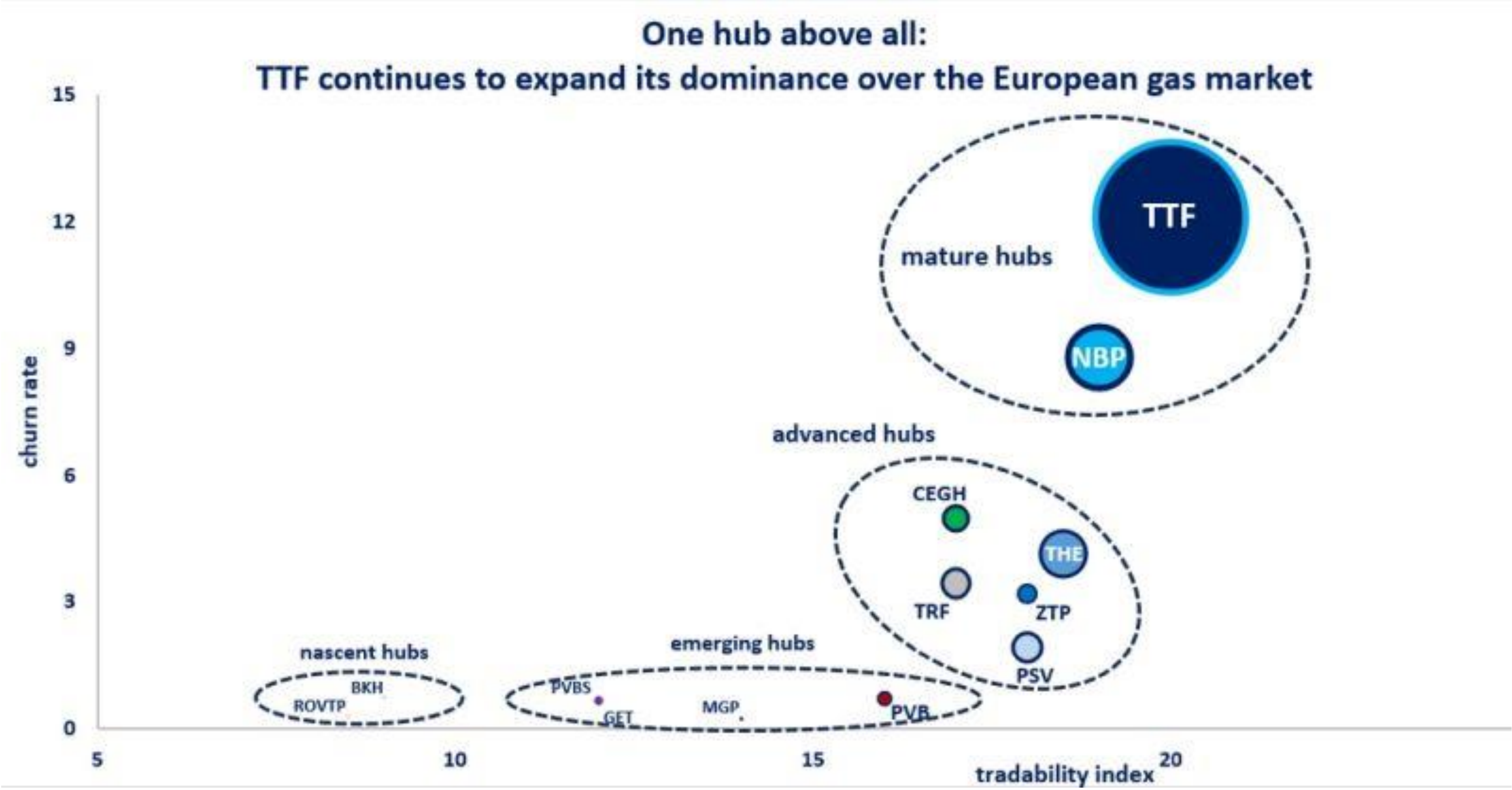
- As long as 2. > 1. then US gas will flow into global market
  - Constrained by the volume of US export capacity

HH Price

- Gas is produced in association with shale oil
  - Economics largely driven by economics of oil
- When oil prices do not support production, associated natural gas production also stops
  - With little to none consideration of natural gas prices
- Less gas production in oil shale plays could mean two things:

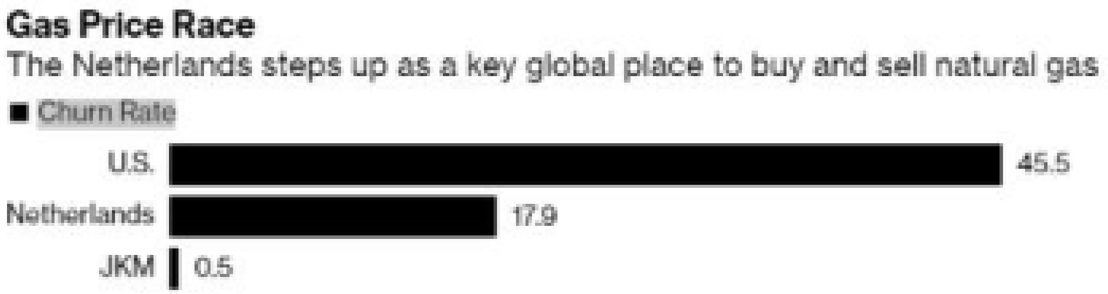
1. Increase in production elsewhere
2. Higher prices

TTF Main Hub

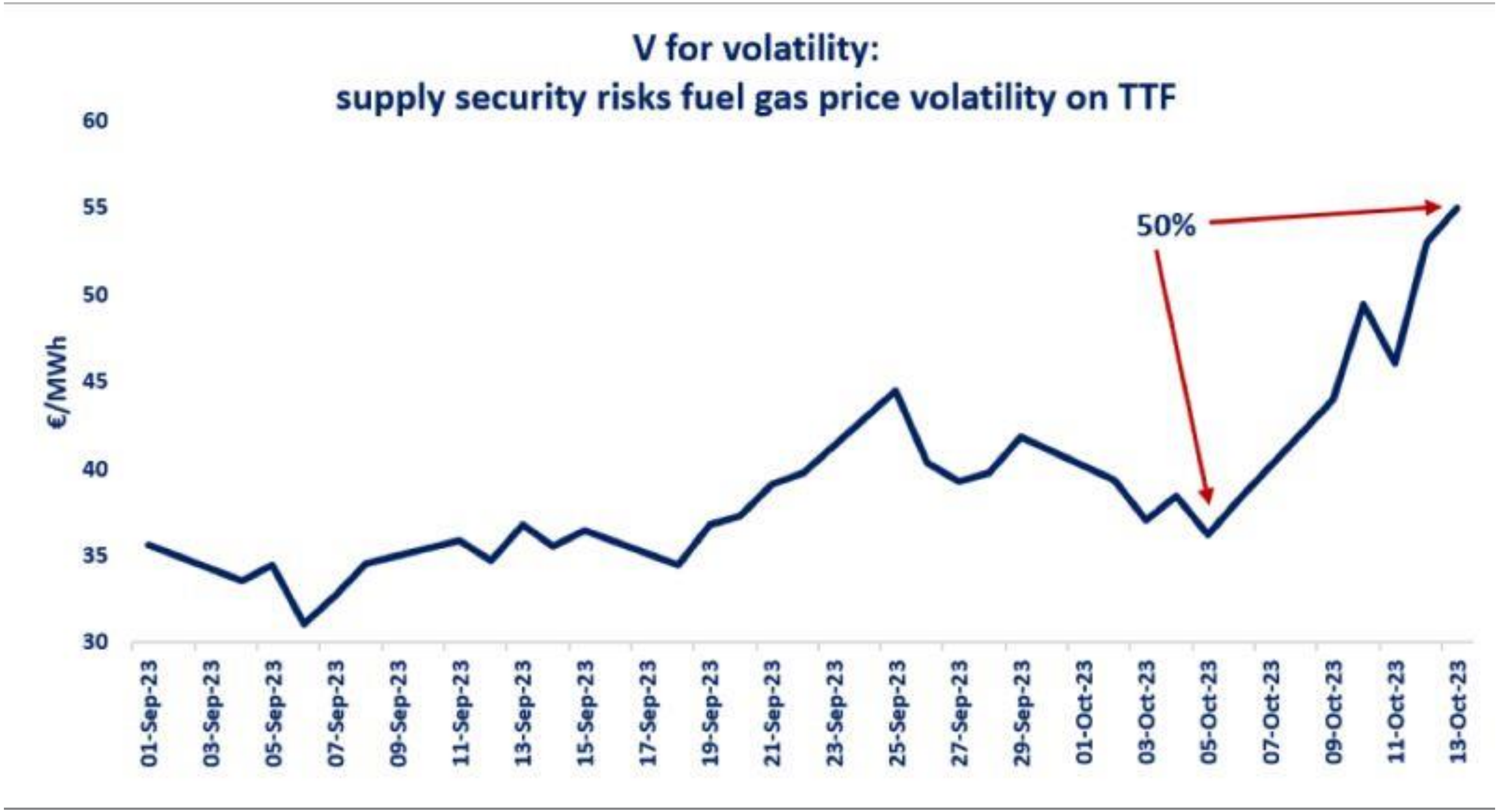


One hub above all:
TTF continues to expand its dominance over the European gas market
churn rate
15
12
9
6
3
0
5
10
15
20
tradability index
mature hubs
TTF
NBP
advanced hubs
CEGH
THE
TRF
ZTP
PSV
nascent hubs
BKH
ROVTP
emerging hubs
PVBS
GET
MGP
PVB

TTF the Global Hub



Volatility TTF



Arbitrage & Freight

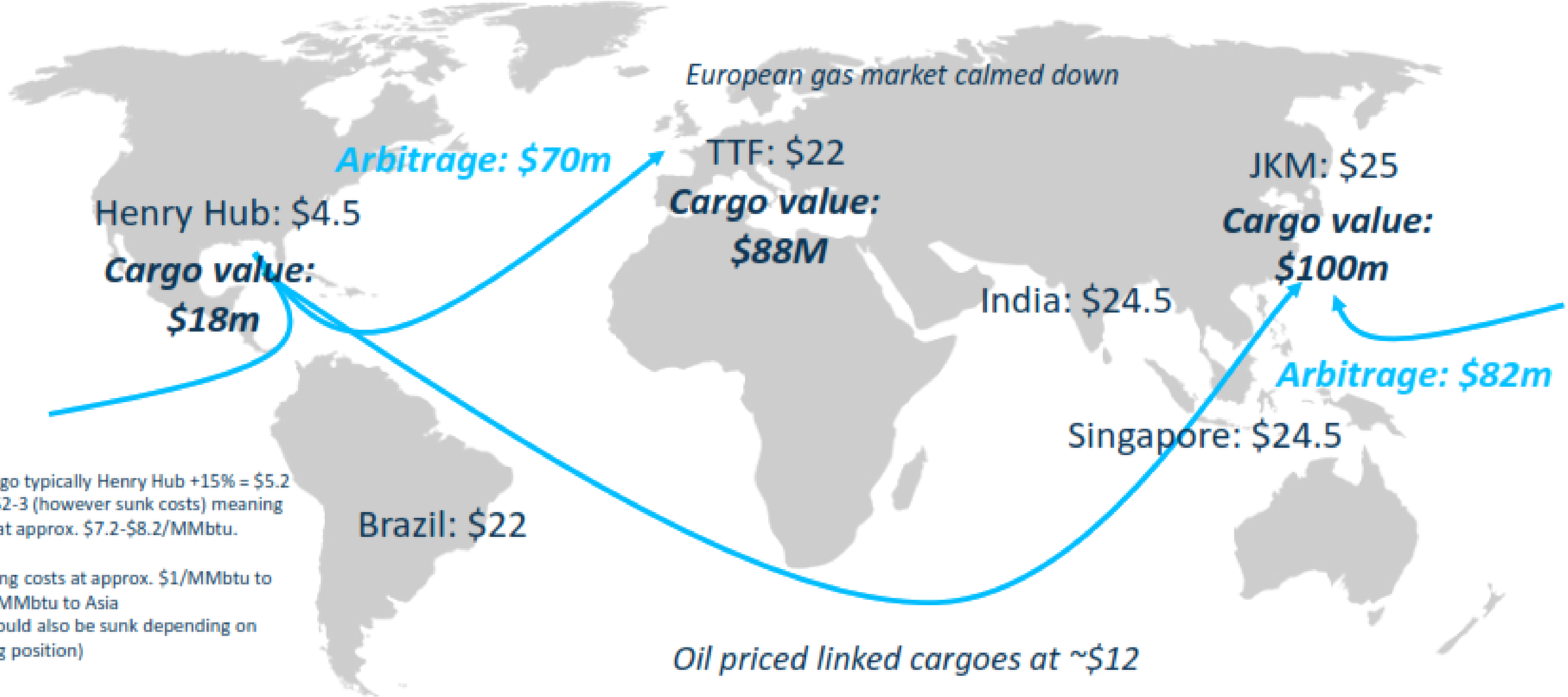
The LNG Product Market on November 30, 2021:

The LNG Product Market on December 21, 2021

Full price US cargo typically Henry Hub +15% = $5.75 + tolling fee of $2-3 (however sunk costs) meaning full costs (FOB) at approx. $7.75-$8.75/MMbtu. Then add shipping costs which could also be sunk depending on trader's shipping position)

1) Source: Platts
2) Assume large cargo of 4trn BTU or approx. 172,000 cbm

LNG Market Feb 15, 2022



Full price US cargo typically Henry Hub +15% = $5.2 + tolling fee of $2-3 (however sunk costs) meaning full costs (FOB) at approx. $7.2-$8.2/MMbtu.

Then add shipping costs at approx. $1/MMbtu to Europe and $2/MMbtu to Asia (shipping cost could also be sunk depending on trader's shipping position)

Case Study Price Arbitrage

- Very low US prices
- Arbitrage calculations based on a FOB price

- For the US markets prices will be calculated on the basis of Henry Hub
- The FOB price will be calculated on the same basis
- Market arbitrage opportunities will be determined in part on differing transport and regas costs

FOB Price	in $/MMBtu
US Price	$1.58
+ Risk Premium (115%)	$0.23
+ Liquefaction costs	$2.25
Price FOB	$4.06

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FOB Price	[illegible]
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+ Risk Premium (115%)	$0.23
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Price FOB	$4.06

Sales Price	[illegible]
Japan (JCC estimate)	$3.85
India (JKM Spot)	$2.08
UK (NBP)	$1.45
Netherlands (TTF)	$1.88

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Japan (JCC estimate)	$3.85
India (JKM Spot)	$2.08
UK (NBP)	$1.45
Netherlands (TTF)	$1.88
Carriage	**in $/MMBtu**
Japan	$1.73
India	$1.25
UK (NBP)	$0.81
Netherlands (TTF)	$0.93

Case Study Price Arbitrage

- Very low US prices
- Arbitrage calculations based on a FOB price

Potential Margin	In $/MMBtu
Japan (JCC estimate)	-$1.94
India Spot	-$3.24
UK (NBP)	-$3.42
Netherlands (TTF)	-$3.11

Case Study Price Arbitrage

- Capacity spot delivery slot for Japan?
- Better to pay Take or Pay rate in US, then actually load cargo?
- When will buyers stop buying?
  - If HH+ USD 2
  - TTF=??
- Assuming no currency risk

Case Study Price Arbitrage

- Cargoes will be lifted until ToP price level
- Even with negative returns

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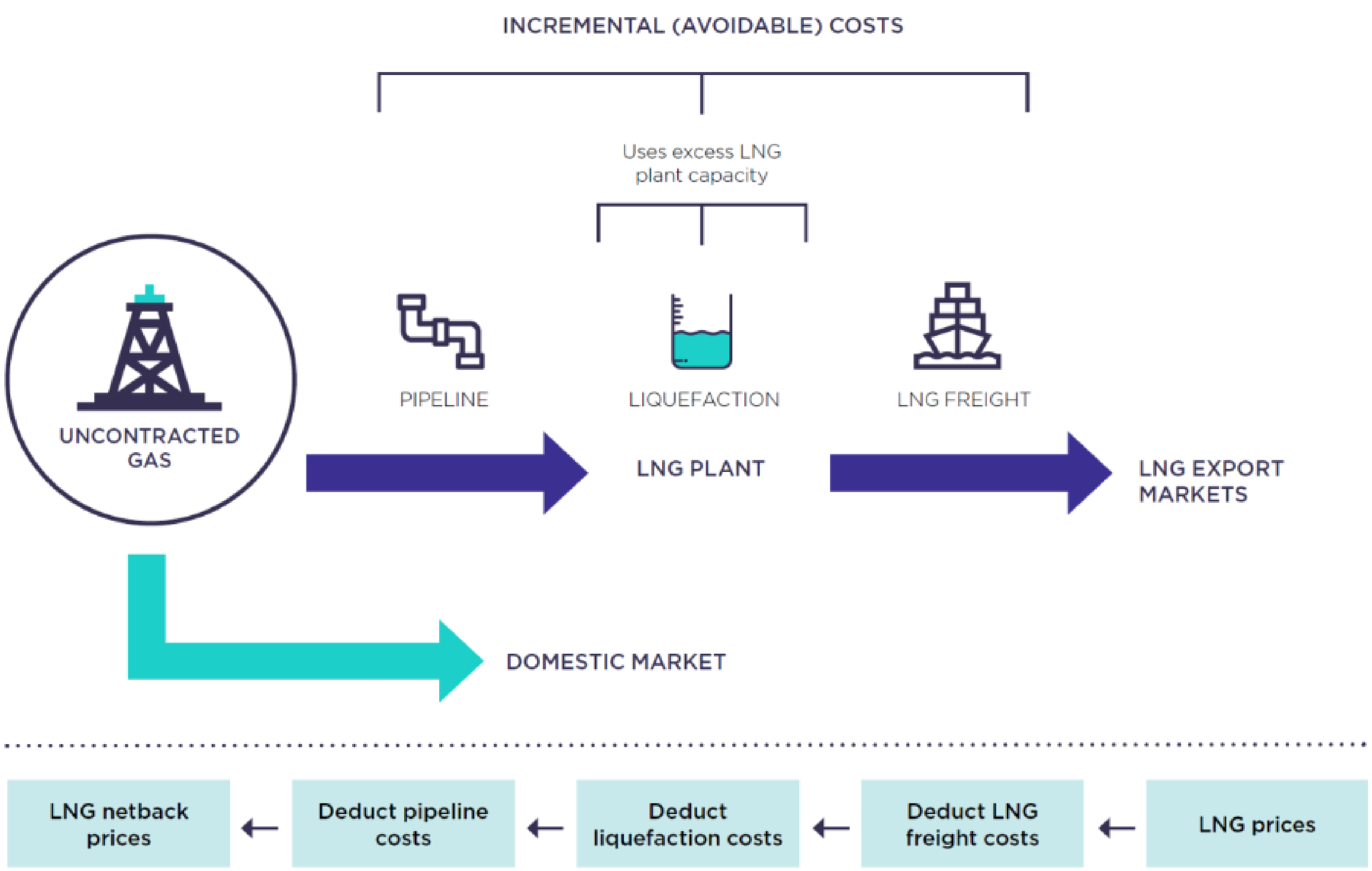
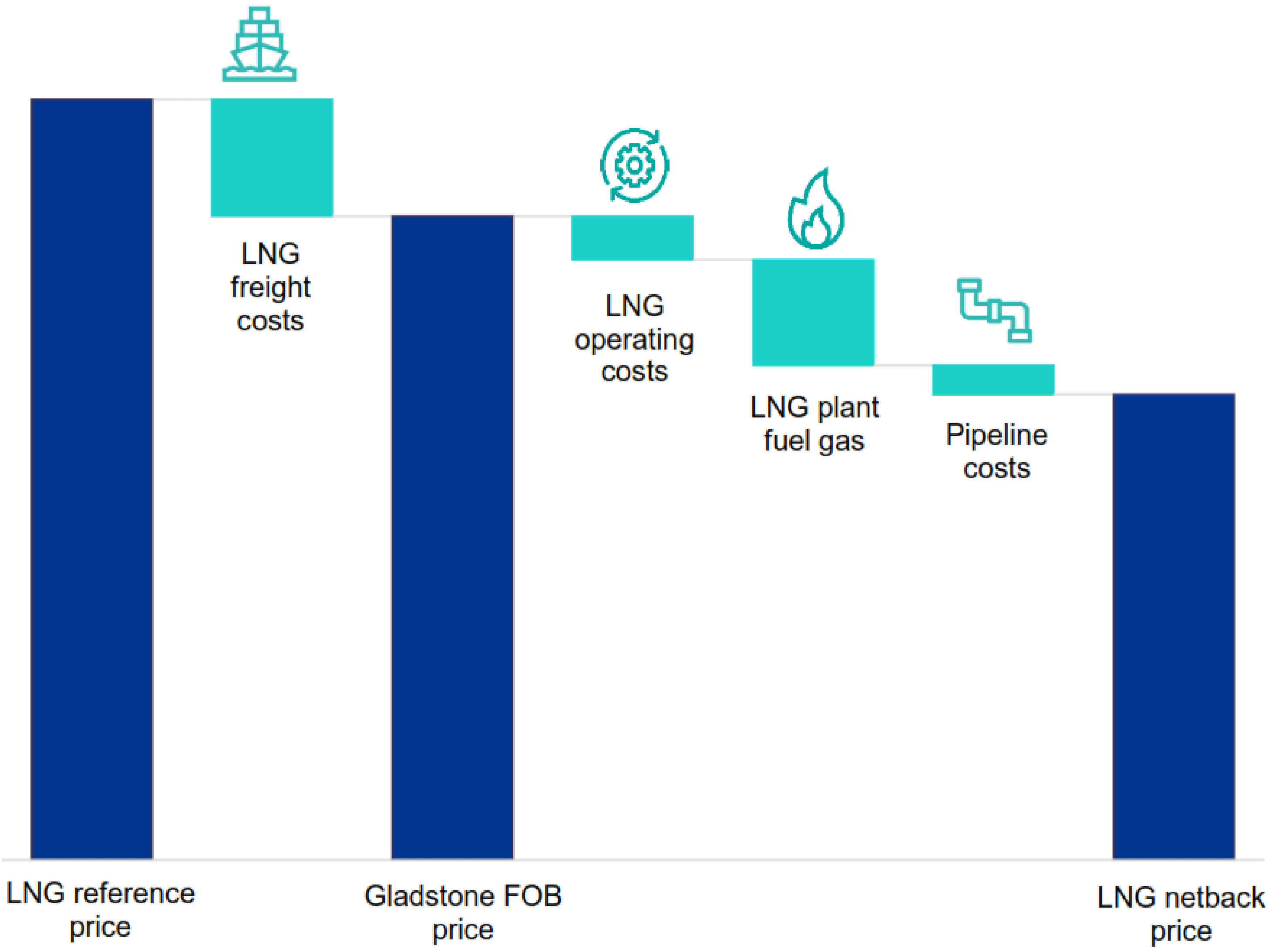
LNG Shipping

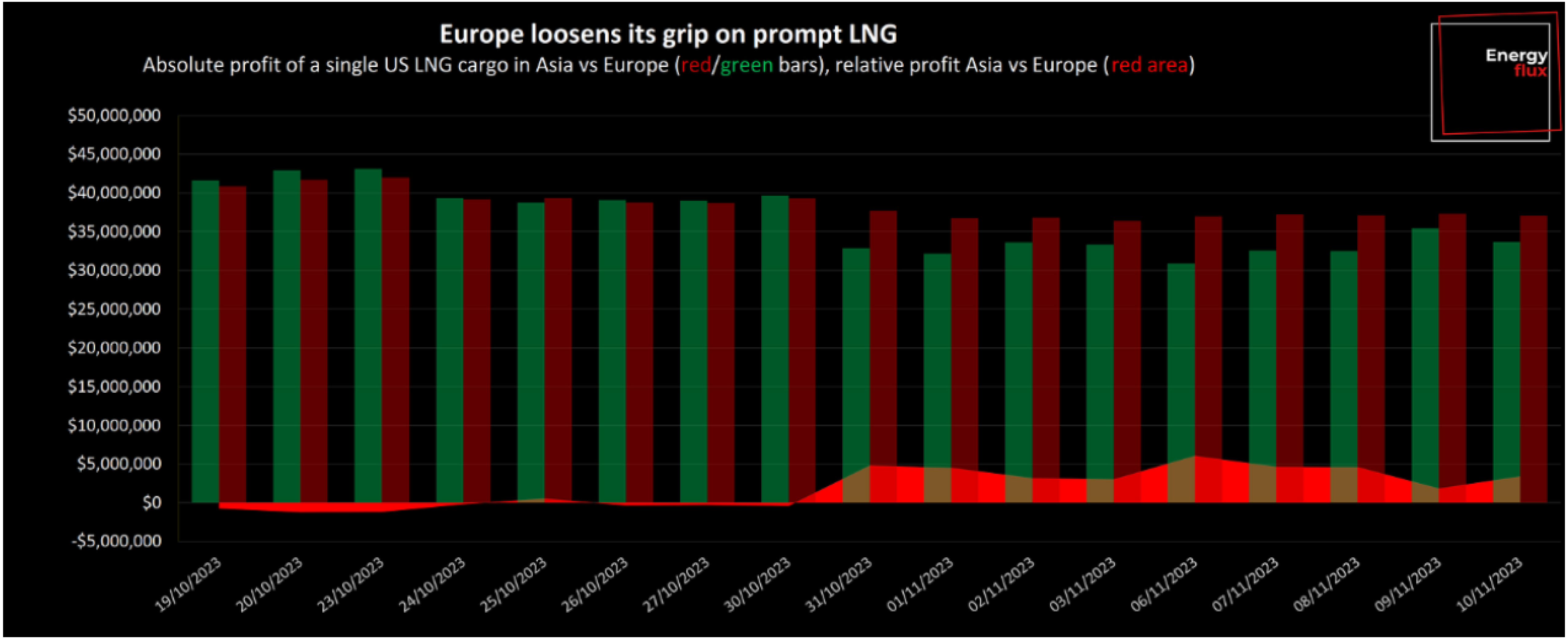
LNG Netback Prices



LNG Netback Prices

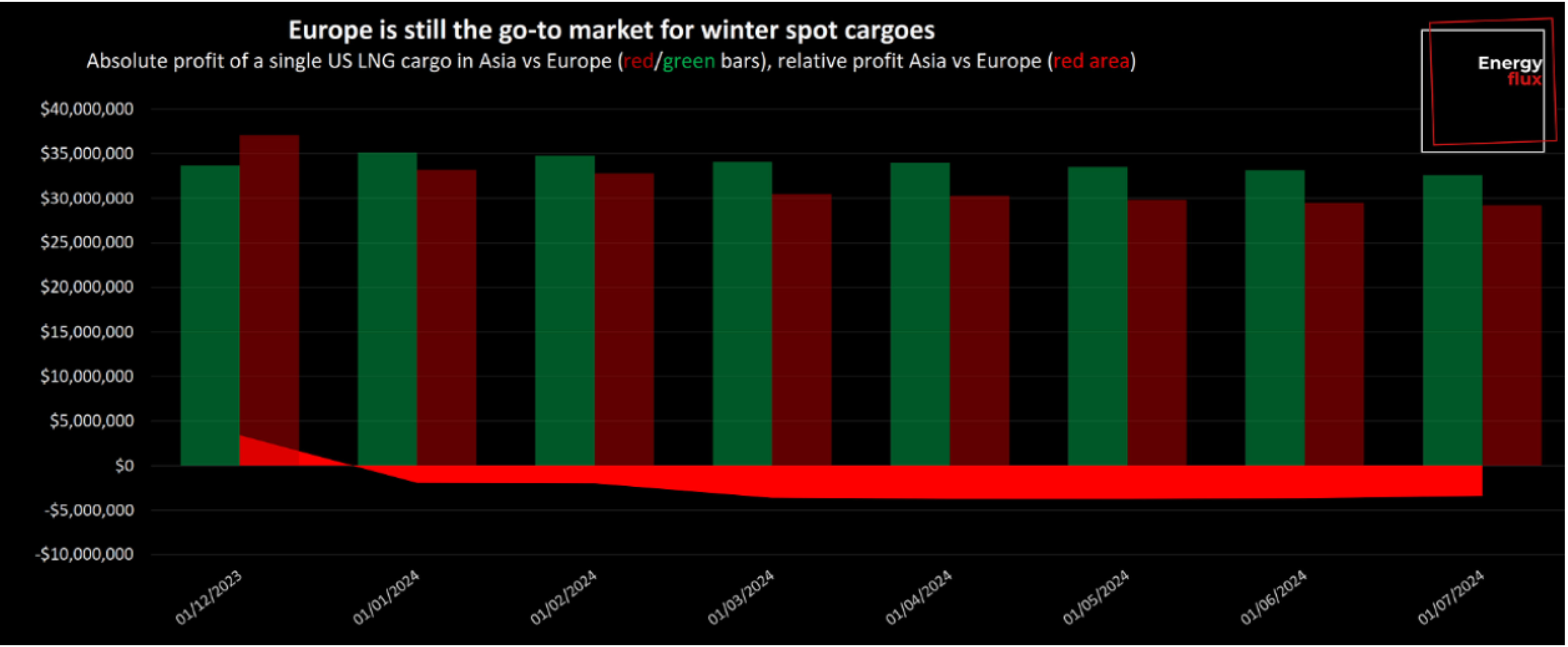


Net Backs



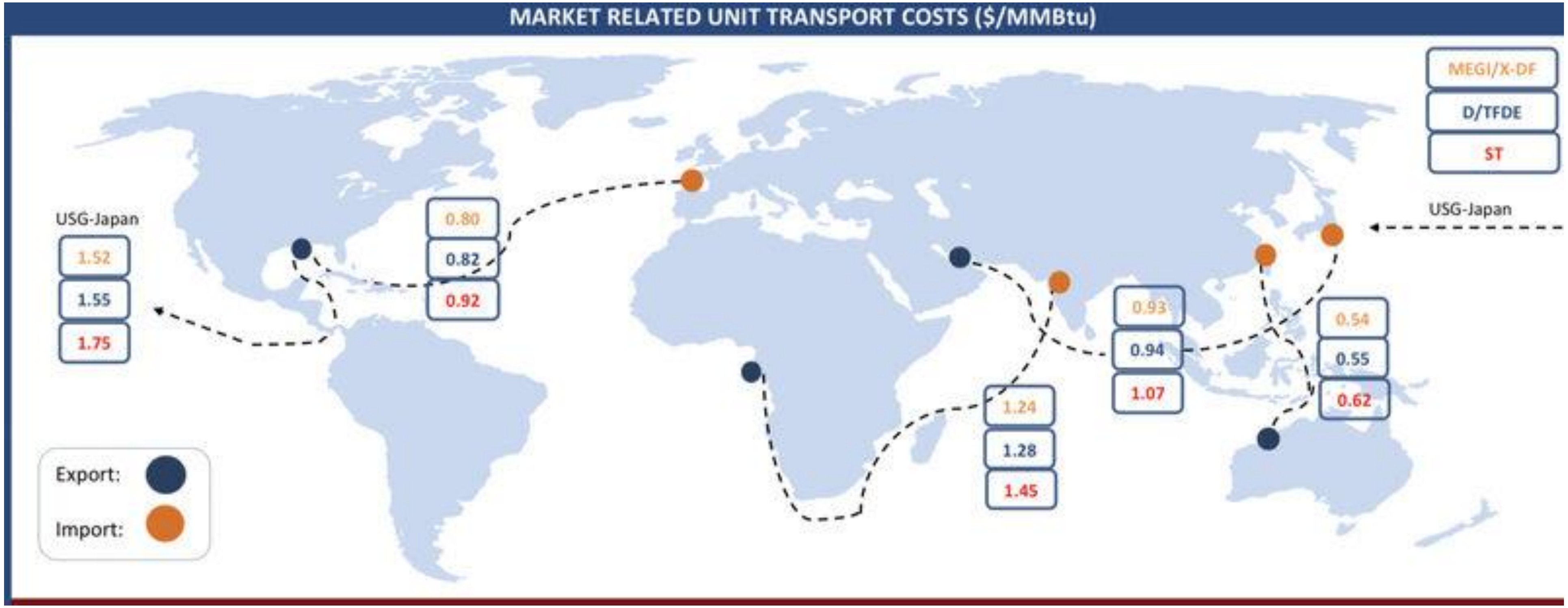
Green bars = Profit per cargo in Europe. Dark red bars = Profit per cargo in Asia. Bright red area = Difference (Asia profit minus Europe profit). Vessel size = 160,000 cm. Profit = TTF/JKM price minus Henry Hub +15% and vessel charter, fuel and regasification (per Capra methodology). Liquefaction is deemed a sunk cost. Data from C ME, XE, Fearnley's. Chart by www.EnergyFlux.news

Net Backs



Europe is still the go-to market for winter spot cargoes
Absolute profit of a single US LNG cargo in Asia vs Europe (red/green bars), relative profit Asia vs Europe (red area)
Energy flux
$40,000,000
$35,000,000
$30,000,000
$25,000,000
$20,000,000
$15,000,000
$10,000,000
$5,000,000
$0
-$5,000,000
-$10,000,000
01/12/2023
01/01/2024
01/02/2024
01/03/2024
01/04/2024
01/05/2024
01/06/2024
01/07/2024

Shipping Costs



Shipping Costs

- Costs of shipping is current spot charter rates plus fuel and canal if any (Time Charter)
- Exporting a cargo from US, where to go?
- Early May: June prices (journey take some time) is Northeast Asia is $9.6/Mmbtu
- If you go to Europe, transportation cost is less, but June TTF is “only” $8.8/MMbtu
  - 0.8/MMbtu less than Asia.
- Difference in transportation costs is $0.7/MMbtu
- These are the calculations cargo owners face when deciding whether to send a cargo to Europe or Asia.

LNG shipping costs

Shipping cost components

- Chartering fee
- 3 ways to secure access to shipping capacity:
  (1) own vessel capacity
  (2) time charter
  (3) single voyage or spot charter

LNG Shipping Costs

- **Brokerage**
  - Vessel charters are typically arranged through specialist brokers and attract a 1-2% fee
- **Fuel consumption**
  - Bunker consumption directly proportional to distance and speed of vessel
    - 2nd largest cost after chartering
  - Most LNG vessels can burn fuel oil, boil-off gas or a blend of both in their boilers.
    - Fuel cost closely tied to boil-off gas

LNG shipping costs

- **Boil-off**

- Natural boil-off occurs at a rate of approximately 0.15% of inventory per day

  – At times boil off is forced above this level to further reduce the fuel oil requirements

- Some modern LNG vessels can re-liquefy boil-off gas, keeping the cargo whole (and allowing the use of more efficient diesel engines)

LNG shipping costs

- **Port costs**
  - Components and level of costs of loading and unloading at ports can vary widely depending on location
  - For example, ports in less stable regions can levy large security charges associated with ensuring the safety of the vessel
- **Insurance costs**
  - Insurance required for vessel, cargo and to cover demurrage (liabilities for cargo loading and discharge overruns)

LNG shipping costs

- **Canal costs**

- Suez canal transit costs complex function of vessel dimensions and cargo (laden voyages being more expensive)

  - LNG vessels entitled to 35% discount
  - Costs USD 300-500k per transit

- Panama canal tariffs

  - Any reduction will increase competitiveness and influence of Henry Hub priced US Exports on Asian pricing

LNG shipping market: Fundamental drivers

- 2 main drivers demand LNG shipping capacity

  1. Volume of LNG to be shipped

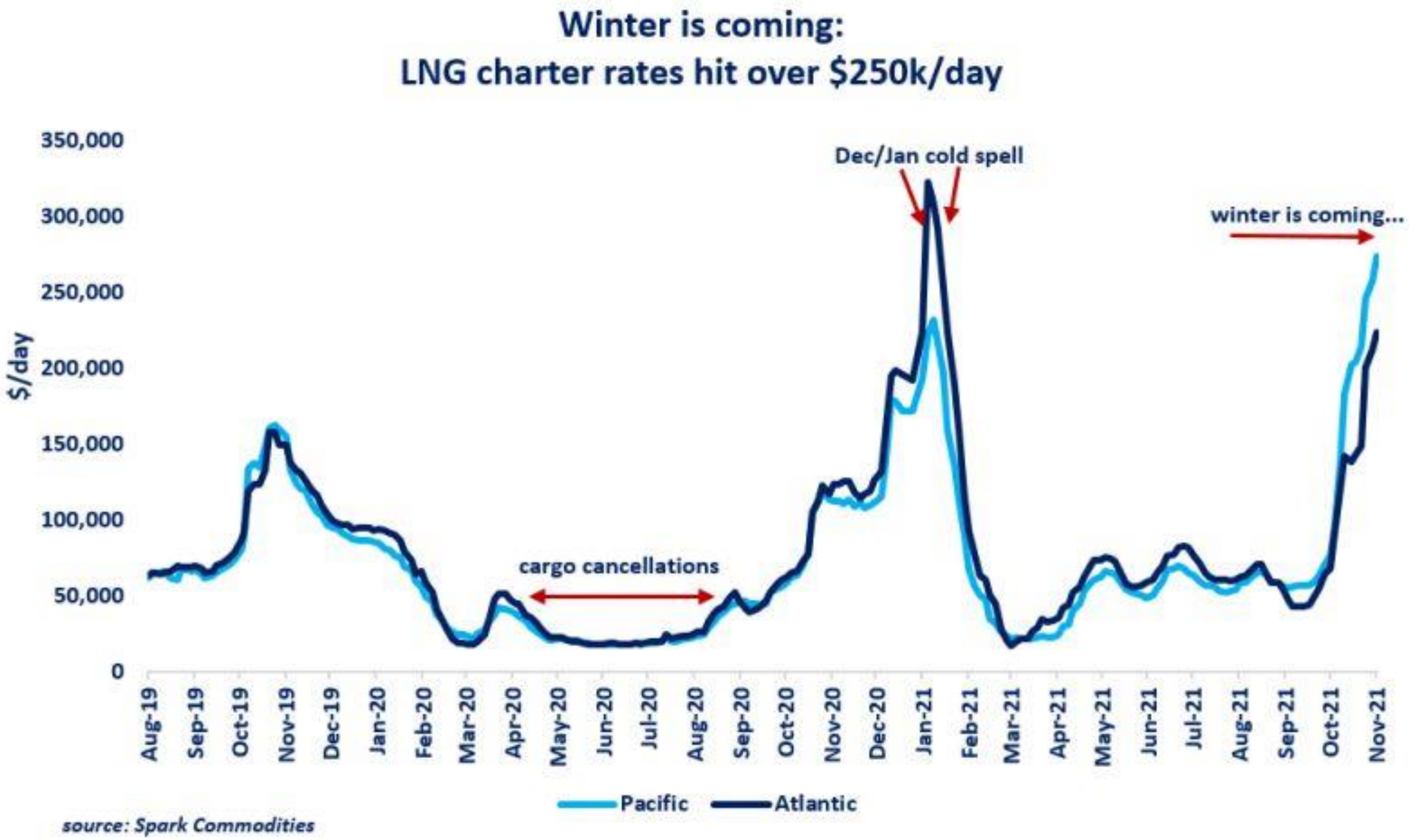
  – Higher LNG demand = higher demand shipping capacity

  2. Avg. journey time & proportion of ballasted (un-laden) voyages

  – Longer average voyages and a higher proportion of ballasted voyages requiring more shipping capacity to move a given volume of LNG

- Understanding LNG trade flows and global LNG demand growth important in understanding tightness in LNG shipping capacity

Charter Rates



Cargo Swap

LNG Freight Derivatives

- 2018- Baltic Exchange launched BLNG indices
  - Methodology includes ballast bonus and/or repositioning fees paid to shipowners

- ***LNG forward-freight agreements (FFAs)***
  - Cleared via CME exchange
  - Traded by over a dozen counterparties
  - Contracts covering periods out to the end of 2021
  - In first nine months of trading, CME has cleared almost 1,000 days of FFA contracts

LNG Freight Derivatives

- Intercontinental Exchange [ICE] launched LNG freight futures contracts based on Spark Commodities’ price assessments
  - Supplement the existing Baltic routes used to settle LNG FFAs cleared and listed by CME
- First trading day 22 March,2021
  - Forward curve going up to and including December 2023
- Spark30S Atlantic
  - Round-trip voyages between the US Gulf Coast and North West Europe
- Spark25S Pacific
  - Between Australia -Japan, Korea, Taiwan and China

LNG Freight Derivatives

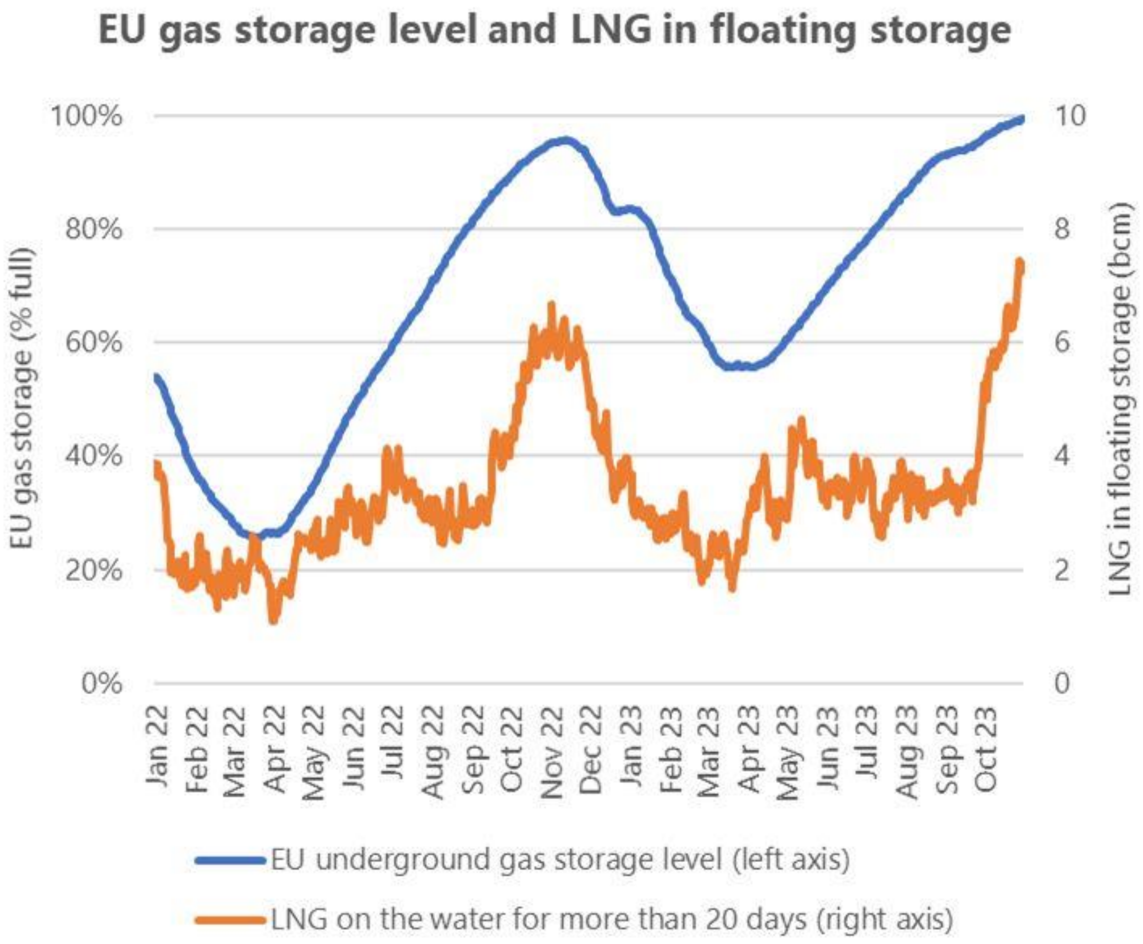
- Allows participants to manage exposure to spot freight rates
- Smaller end-users didn’t want to buy Des LNG from traders
  - Now commit to FOB cargoes having covered any freight risk in advance
- Shipowners can look to balance utilization and price risk
  - Combination of long-t erm charters and hedged spot trading

FSRU Germany

FSRU

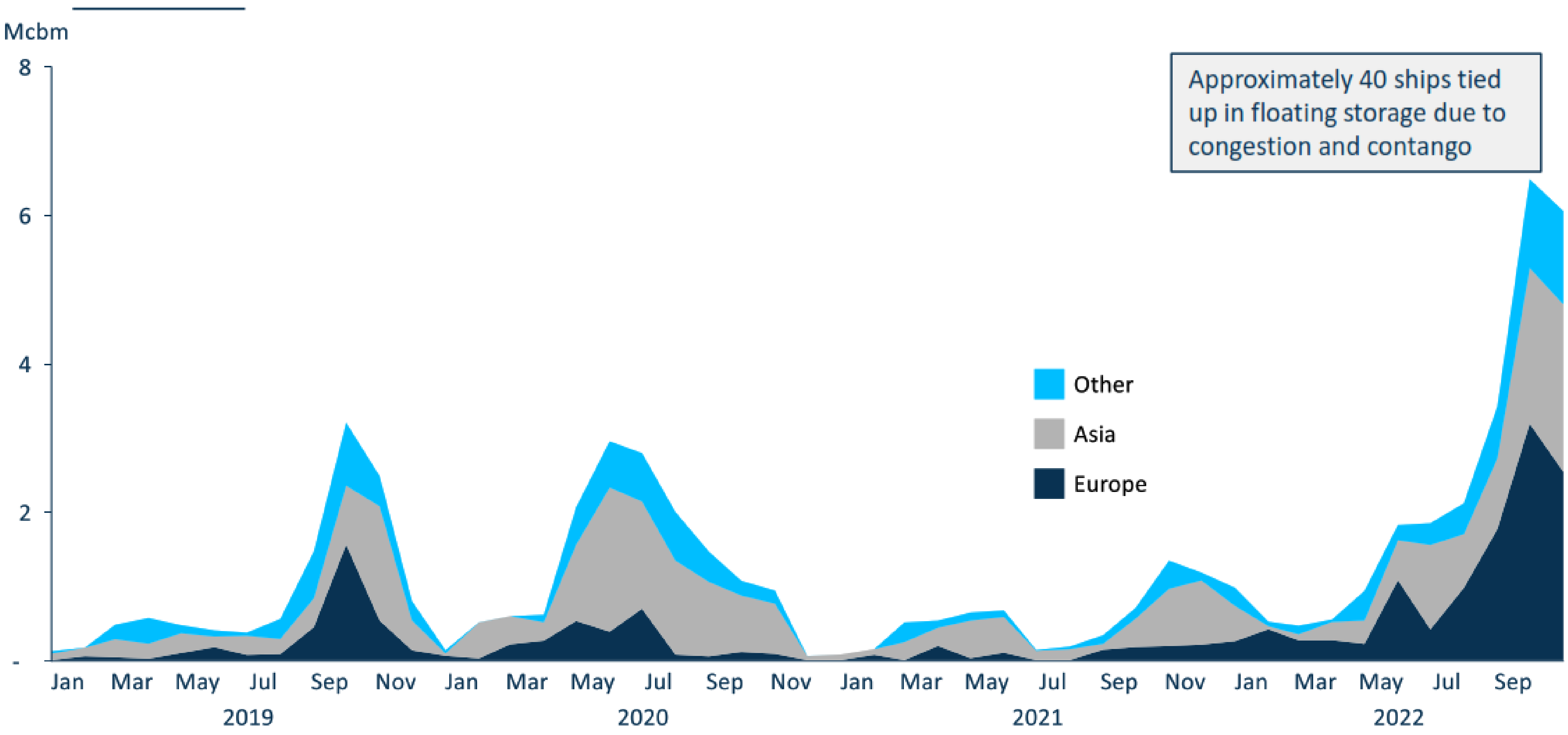
- TotalEnergies
- German Baltic Sea
- Maximum capacity of 5 billion cubic meters a year
- More than 5% of Germany’s annual consumption
- Will start transporting 4.5 billion cubic meters a year from Dec. 1
- Russian gas now makes up app. 25% of Germany’s total gas imports
- From 45% in mid-May

Floating Storage LNG



Source: GIE Aggregated Gas Storage Inventory, Bloomberg

Floating Storage Record Impact Spot Market



Floating Storage

Fall 2018

- Anticipation surge Chinese demand and prices
- Mild winter damped LNG demand
- Forcing traders to discharge LNG and release the vessels
- Causing a crash in spot charter rates
- Gamble as weather-driven demand is unpredictable

Floating Storage

- Price signals indicate option to store LNG in vessels later this year is again becoming attractive
  - In both Asia and Europe
- Price premium for later contracts, or contango, is building up for both spot Asian LNG prices and NBP/TTF
- Leaving traders weighing how to profit should they choose to store gas at sea

Floating Storage

- Lot of floating LNG storage from September
- Could also tighten shipping rates before winter
- Price gap
- Estimated storage cost of 0.60-0.75 /mmBTU a month
- Current freight rates allow for floating storage opportunities in September, October and November

Floating Storage

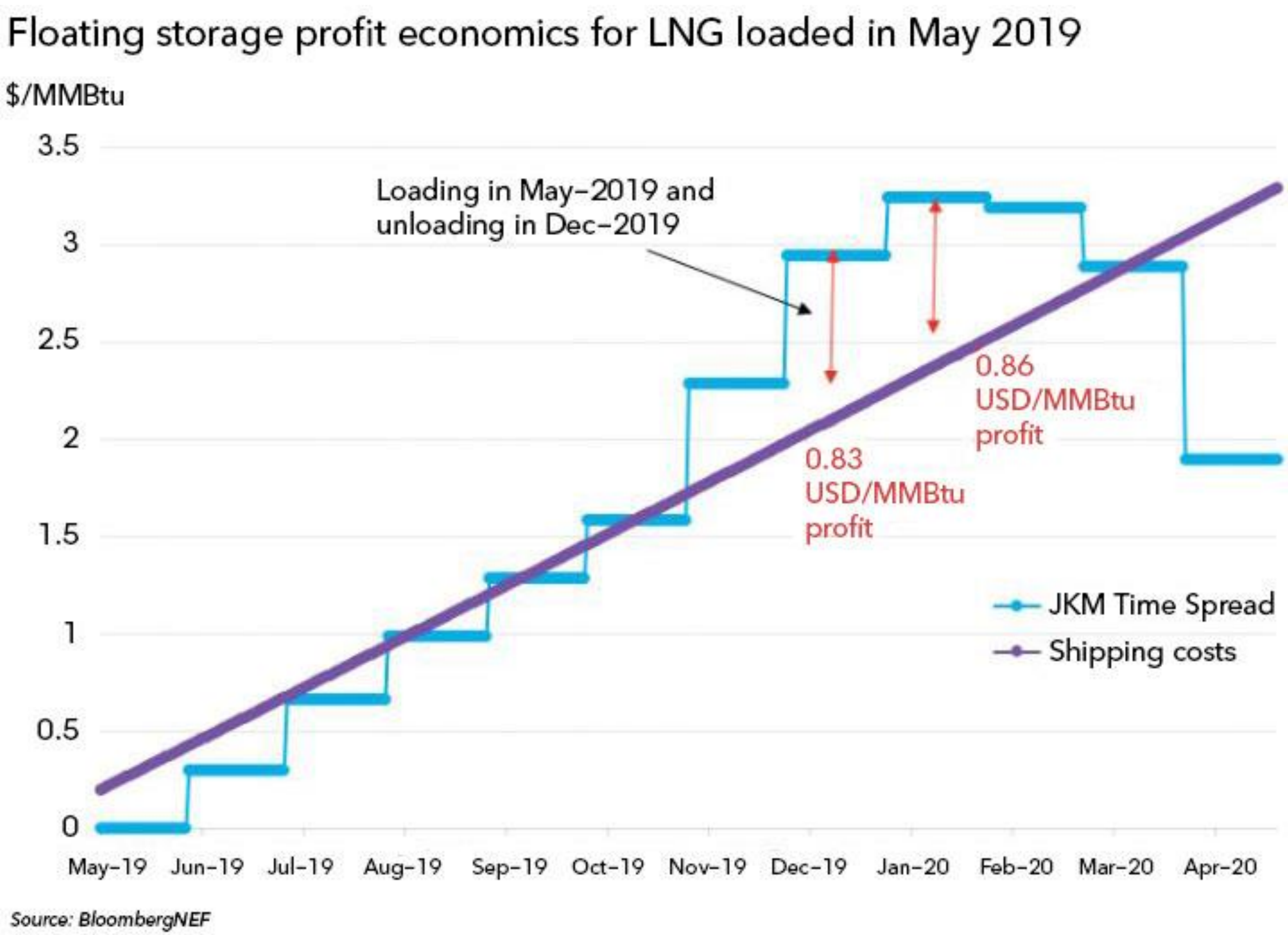
- JKM Dec futures app. $2.30 per million Btu higher than September contracts
- Premium of $7.6 million for a 3.3 trillion Btu cargo
- Actual profit strongly dependent on costs :
  - Renting a tanker
  - Keeping the fuel in liquid form at minus 162^{o}C
  - Newer vessels with low levels of boil off and/or reliquefaction plants are the most suitable candidates for such trade

Floating Storage

Short-cycle floating storage for LNG loaded in September 2018

Source: BloombergNEF

Floating Storage



LNG Contracting

LNG Contracts

Long term

- “Take-or-pay” clauses
- Some contracts fixed price for whole duration
- Others price renegotiated if market conditions change

Short Term

- More flexibility

4 Key Trends LNG Contracting

1. Reduction in average contract length

- Financing new LNG supply projects traditionally with long term, oil-indexed contracts
- Model increasingly challenging for LNG buyers
- Growing penetration of hub pricing
  - Mismatches LT contract prices and hub prices increasingly difficult to pass through to customers or to absorb into supplier portfolios
- Buyers in Asia, Europe and Latin America are pushing for shorter and more flexible gas linked contracts

4 Key Trends LNG Contracting

2. Growing importance of portfolio players

- Trading companies focus LNG midstream flexibility
  - Portfolios built around ST contract positions and access to shipping, regas and storage capacity
- Oversupply
  - Increases availability of cargoes to trade
  - Strengthening role hub prices for managing portfolios
    - Access to flexibility key
    - Flexibility priced, hedged and optimized based on liquid hub price signals

4 Key Trends LNG Contracting

3. Declining buyer credit quality

- Decline in the credit ratings of LNG buyers
  - Traditional A-rated buyers (e.g. Japan, Korea) are currently relatively well contracted from LT deals
  - Stronger demand growth from emerging buyers
    - India
    - Egypt
    - Pakistan
  - Reduces credit quality
  - Higher buyer credit risk; shorter contract lengths

4 Key Trends LNG contracting

4. Increasing penetration of hub prices

- Strong buyer preference hub indexed LNG contracts
- Converged global gas pricesgrowing confidence in Atlantic basin spot price signals (NBP, TTF, Henry Hub) as a global benchmark for LNG contracting
- Spot price signals key driver of LNG flow and optimization decisions for large portfolio players
    - Cargo diversions and cargo swaps

Deconstructing Flexibility

Source of flex	Description	Examples
1. Pricing flex	Choice of price indexation. Complex price formulas with embedded sold / held optionality.	• Oil • S-Curves • Caps / floors, index options
2. Commodity flex	Flex to lift / deliver different absolute commodity volumes.	• Upwards Quantity Tolerance (UQT) • Downwards Quantity Tolerance (DQT) • Cargo volume tolerance • US toll cargo lift flexibility
3. Locational flex	Flex to deliver cargos to different locations.	• Diversion rights e.g. o FOB to deliver anywhere subject to rotations o DES right to deliver within a fixed subset of locations e.g. North West Europe + Spain
4. Timing / intertemporal flex	Flex to move volume into different periods.	• Annual Delivery Program (ADP) timing flex • Delivery window flex
5. Portfolio / logistical flex	Broad range of inherent flex in optimisation of cargo and vessel logistics.	• Cargo matching • Floating storage • Vessel allocation & operations (e.g. route, speed, propulsion)

- Flexibility often held by buyer & seller

ICE

Flexibilities in Contracts

Timing

- ADP: Annual delivery programme with each counterparty
- Arrival windows
- Penalties for arriving outside the window

Volume

- Loaded and discharge volume flexibilities per cargo
- Annual volume tolerances on purchase and sales contracts
- Additional cargoes nomination at some time notice

Flexibilities in Contracts

- ***Shipping***

- Vessel choice
- Speed and route
- HFO versus Gas

Prices & Markets

- Destination and pricing formula
- Sub-chartering versus LNG trading

LNG contracting structures

1. Master LNG SPA ("**Master Agreement**")

   – All standard terms and conditions of LNG trading
   – No firm obligation on seller to sell or on buyer to buy

2. Confirmation notice/memorandum ("**Confirmation Notice**")

   – Once signed, Confirmation Notice together with Master Agreement ;valid binding contract

Master Agreements

- 3 international models:

- **EFET**
  - European Federation of Energy Traders
  - DES terms. Generally considered pro-trade

- **GIIGNL**
  - International Group of LNG Importers
  - DES model and FOB model

- **AIPN**
  - Association of International Petroleum Negotiators
  - One model, FOB and DES

Legal impact Covid-19

- Market disruptions
  - Notices of force majeure
  - Requests for cargo diversions
  - Delays
  - Cargo cancelations and deferrals

- US-style LNG cargo cancellation clauses to become more common
- Flexibility to renegotiate will attract buyers to sign long-term deals
  - Contract reopening mechanism, allowing for a comprehensive contract adjustment

Legal impact Covid-19

- Asian price review clauses
  - Traditionally at narrowly defined time intervals
    - Every five or 10 years
  - More and more triggered by the notion of "change of circumstances"
    - In European price reviews